

Carry distance, dispersion and gapping

A statistical profile of Michiel's bag from 211 logged strikes across 6 range sessions, cross-checked against 598 GPS-tracked strokes (putts excluded) from 14 rounds of play.

TIGHTEST CLUB (N > 20)

4.4%

9 iron · $s = 5.5$ m
on 26 strikes

LOOSEST CLUB

20.1%

5 iron · $s = 26.6$ m
spans 81-168 m

DISPERSION CONE

5.3°

angular SD of the
carry line, whole bag

MID-IRON PUSH

+11 m

7 iron mean offline
 $p < 0.001$, $n = 31$

The bag, as measured

Carry is the number to club off; the 80% window is where four shots in five actually finished. Roll and total are medians on range turf in cool April air, and will run further on a firm summer fairway - see page 12 for the on-course check.

CLUB	N	CARRY	95% CI ON CARRY	80% CARRY WINDOW	SPREAD s	CV	ROLL	TOTAL
Driver	29	177 m	166 - 188	147 - 198 m	19.1	! 10.9%	+17	203
5 Wood	6	151 m	139 - 175	139 - 175 m	17.6	! 11.3%	+15	189
5 Iron	17	137 m	109 - 153	98 - 158 m	26.6	x 20.1%	+26	175
6 Iron	8	144 m	137 - 155	137 - 157 m	8.8	! 6.0%	+17	159
7 Iron	31	135 m	132 - 140	125 - 149 m	9.3	! 6.9%	+14	150
8 Iron	35	128 m	123 - 130	115 - 141 m	14.3	! 11.4%	+12	141
9 Iron	26	123 m	120 - 126	116 - 130 m	5.5	+ 4.4%	+9	136
Pitching Wedge	8	115 m	112 - 117	111 - 118 m	4.0	+ 3.4%	+7	124
Gap Wedge	13	93 m	87 - 99	85 - 102 m	7.0	! 7.5%	+10	101
Sand Wedge	17	82 m	79 - 85	75 - 89 m	6.1	! 7.4%	+10	92
Lob Wedge	8 + 13	28 / 48 m	partial shots	to 30 m / 50 m targets	5.3 / 4.7	! 19 / 10%	—	—

What the numbers say

• The 5 iron is not a 5 iron.

Its median carry (137 m) is 7 m SHORTER than the 6 iron's, and its spread ($s = 26.6$ m) is three times the 6 iron's. It is the only inverted rung in the bag.

• Two rungs of the ladder are broken, and four more are marginal.

5i/6i and 8i/9i overlap so far that the longer club wins under 65% of shot pairs; Dr/5W, 5W/5i, 6i/7i and 7i/8i all sit in the 0.75-0.78 band. Effective gapping is seven clubs wide, not ten.

• Direction is a bigger leak than distance.

Offline spread scales at 5.3° of carry - ± 21 m with the driver, before any bias at all. The 7 and 9 irons carry a significant push on top of that.

• Day-to-day drift is small; strike quality is everything.

Between-session variance is under 5% of the total for five of seven repeated clubs, and clubhead speed explains under half of the carry spread in the scoring irons.

Data, exclusions and method

Every number in this report comes from 211 usable strikes logged over six range sessions between 11 and 19 April 2026. This page states what was kept, what was dropped, and which estimator is used where - so any figure can be reproduced or contested.

The sample

SESSION	SHOTS	CLUBS	AIR DENSITY	TEMP	RH	SESSION NOTE
Sat 11 Apr 2026	33	3	1.193 g/L	21.1 °C	43%	Driving range na 18 holes. Alles onder 8 was steady
Sun 12 Apr 2026	27	2	1.223 g/L	13.9 °C	47%	Driving range na 13 holes. Big draws
Tue 14 Apr 2026	69	5	1.219 g/L	15.0 °C	39%	Driving range sesh: baby draw
Thu 16 Apr 2026	20	2	1.228 g/L	12.8 °C	62%	Normal driver
Fri 17 Apr 2026	25	3	1.213 g/L	16.1 °C	63%	—
Sun 19 Apr 2026	37	4	1.218 g/L	15.0 °C	55%	—

Exclusions and stratification

• **1 shot removed - radar artefact**

A “Sand Wedge” logged at 213.6 km/h of club speed and 226.7 km/h of ball speed, carrying 147 m. That club speed is 52 km/h above the fastest driver swing in the file, so it is a sensor fault rather than a mishit. It is the only shot removed.

• **21 lob-wedge shots split out, not pooled**

Both LW blocks were hit to a nominated target (“Goal: carry 30”, “Goal: carry 50”). They are two distance-control samples with different means by design; pooling them would manufacture a bimodal “distribution” that describes nothing. They are analysed on page 11 and excluded from the gapping ladder.

• **0 mishits removed - deliberately**

The thinned 8 iron (63.9 m carry off a normal 167 km/h ball speed at a 4.3° launch), the chunked 5 iron (smash 1.00) and the skied driver (7 680 rpm) all stay in. They are strikes you will repeat on the course, and a yardage built on your good swings only is a yardage that is short when it matters. Where they distort a moment, the robust twin is reported beside it.

Estimators used, and why

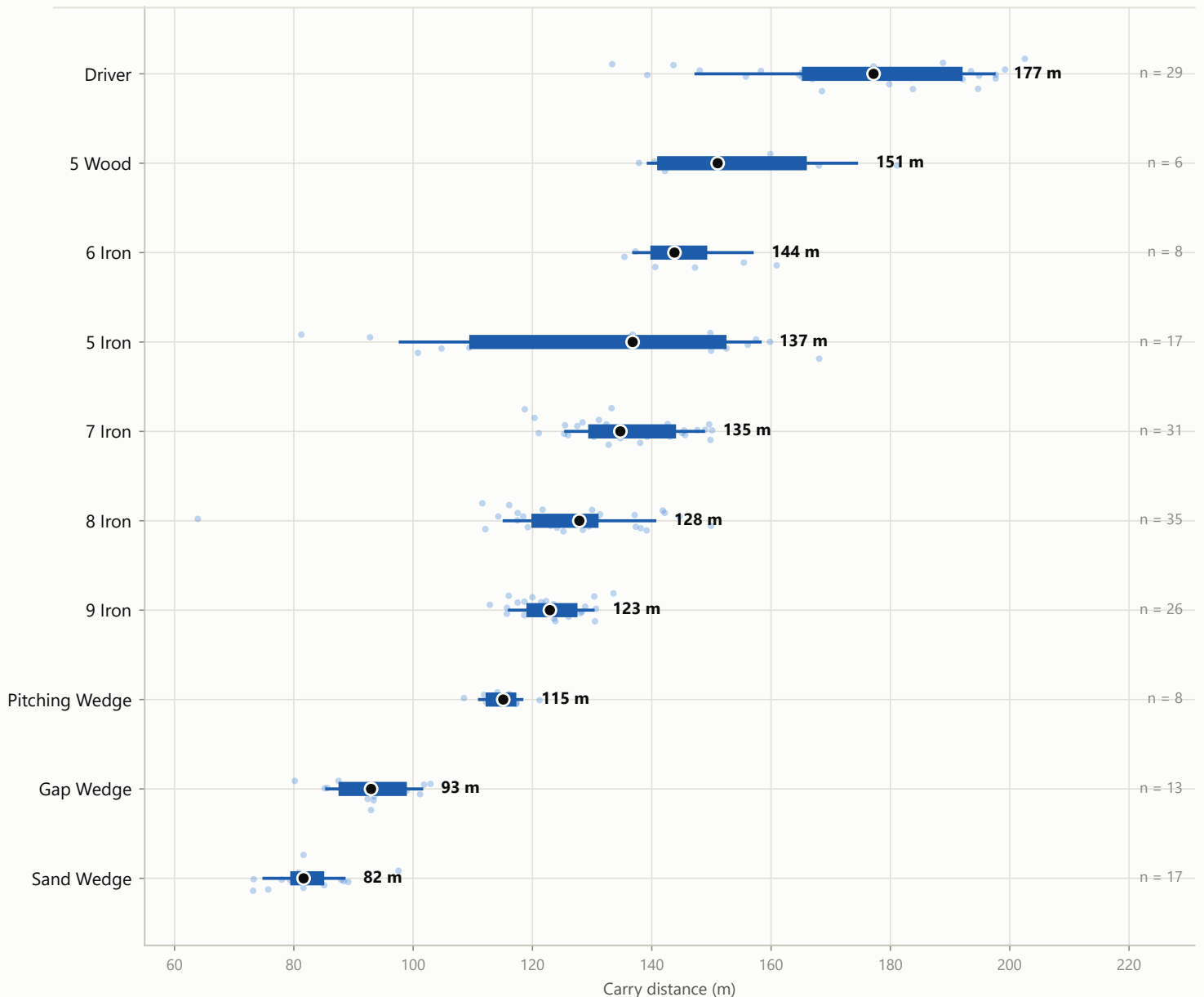
QUANTITY	ESTIMATOR	RATIONALE
Typical carry	Median; H-L and a 10% trimmed mean alongside	Carry is left-skewed: a mishit loses 40 m, a flush one gains 8.
Spread	SD and MAD-based SD (1.4826 × MAD)	SD far above MAD-SD means the tail is wide, not the body.
Uncertainty	Non-parametric bootstrap, 20 000 resamples	No distributional assumption, and honest at n = 8.
Planning window	Empirical p10-p90 and an 80/90 tolerance interval	It covers 80% of future shots with 90% confidence.
Club separation	P(A > B) over all shot pairs, plus Mann-Whitney U	“How often does the longer club win” is the real question.
Two-way dispersion	50% and 90% covariance ellipses (chi-square, 2 df)	Carry and offline error are correlated in the wedges.
Day-to-day drift	Random-intercept model on session (REML), with ICC	Separates “the club changed” from “that swing was poor”.
On-course distance	Two-component Gaussian mixture on GPS shot length	Course data mixes full swings with layups and chips.

Three caveats worth carrying through the report

- Carry is modelled, not measured. The R10 measures ball speed and launch, then flies the ball through a model. Spin was estimated rather than measured on 165 of 211 strikes, so part of the carry spread here is the model's sensitivity to that estimate rather than ball flight. Treat it as an upper bound on true dispersion.
- n is small for half the bag. Five clubs have fewer than 18 strikes. A spread estimated from 8 shots has a 95% interval running from 0.66× to 2.04× the number you computed - a threefold range. Page 5 shows which conclusions survive that and which do not.
- Range air was cool and dense. Session air density ran 1.19-1.23 g/L at 13-21 °C. Warm summer air is nearer 1.16 g/L, worth roughly +2% of carry - part of the on-course gap on page 12, and not separable there from turf roll.

Where every shot finished

One row per club, ordered by median carry. The heavy bar is the interquartile range, the thin whisker the 10th to 90th percentile, and every logged strike is drawn behind it. The width of a row is the club's honest yardage - not the number stamped on its sole.



Reading the rows

- **The 5 iron row is four times as wide as the 6 iron row, and sits to the left of it.**

Two clusters are visible inside it - the 11 April session (median 109 m) and the 16 April session (median 148 m). Page 10 tests whether that is a day effect or a strike effect.

- **The 8 iron's left whisker reaches 63.9 m while its box is only 11 m wide.**

That single thinned strike doubles the club's SD. It is why this report carries a robust spread beside the classical one everywhere, rather than choosing between them.

- **The 9 iron and pitching wedge rows are narrower than the gaps around them.**

Those two clubs are trustworthy to the metre. Nothing else in the bag is, and much of what follows is an account of why.

- **Wherever two boxes share ground on the x axis, the clubs are interchangeable.**

Between the 5 and 6 irons, and between the 8 and 9 irons, the overlap is most of the box. That is the gapping problem, quantified on page 7.

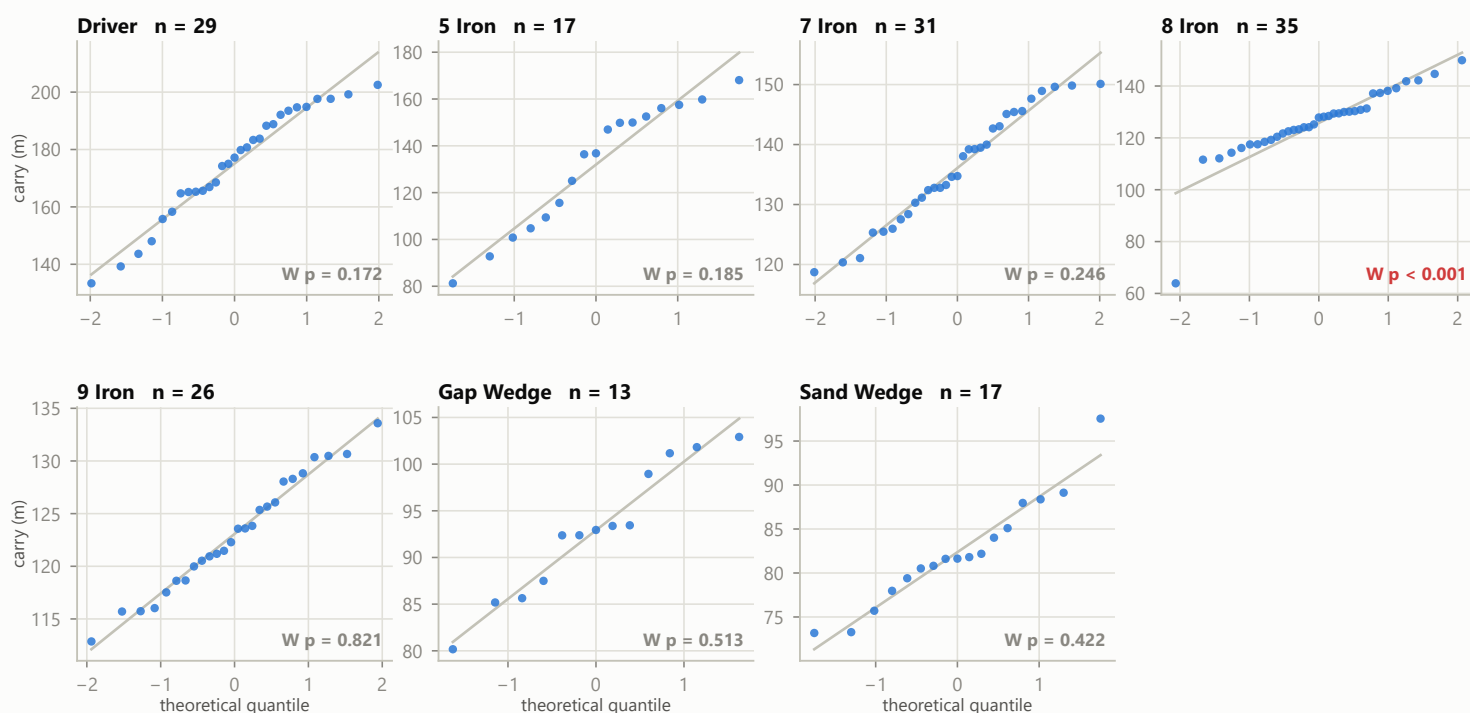
The shape of each distribution

Whether a club's carry is normal decides which interval you may quote. One of the ten clubs fails the test, and it fails through the left tail - which is exactly the tail that matters when the question is whether a shot will carry.

CLUB	N	MEAN	MEDIAN	H-L	TRIM 10%	s	MAD-s	s / MAD-s	SKEW	EX. KURT	SHAPIRO-WILK p	NORMAL?
Driver	29	175.1	177.2	176.5	176.1	19.1	18.5	1.03	-0.53	-0.54	0.1722	yes
5 Wood	6	154.9	151.0	154.3	154.9	17.6	17.6	1.00	+0.57	-1.44	0.3279	yes
5 Iron	17	132.0	136.8	132.3	132.9	26.6	30.7	0.87	-0.51	-1.04	0.1846	yes
6 Iron	8	145.6	143.8	144.7	145.6	8.8	7.4	1.19	+0.80	-0.21	0.5011	yes
7 Iron	31	136.1	134.7	136.2	136.4	9.3	11.8	0.79	-0.13	-1.01	0.2455	yes
8 Iron	35	125.8	127.9	126.7	126.8	14.3	11.0	1.30	-2.22	+9.62	<0.0001	no
9 Iron	26	123.1	122.9	123.1	123.0	5.5	6.4	0.86	+0.08	-0.80	0.8213	yes
Pitching Wedge	8	114.8	115.1	114.8	114.8	4.0	3.8	1.05	-0.02	-0.15	0.9617	yes
Gap Wedge	13	92.9	92.9	92.9	93.2	7.0	8.9	0.79	-0.17	-0.73	0.5129	yes
Sand Wedge	17	82.4	81.6	81.8	82.0	6.1	5.1	1.19	+0.70	+1.12	0.4218	yes

Normal quantile plots

Clubs with at least 13 strikes. Points on the line are normal; a hook down at the left is a mishit tail.



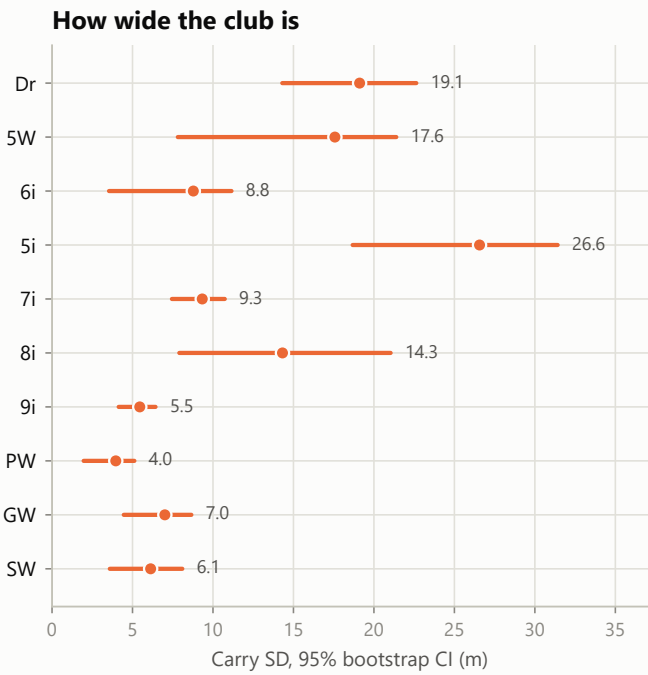
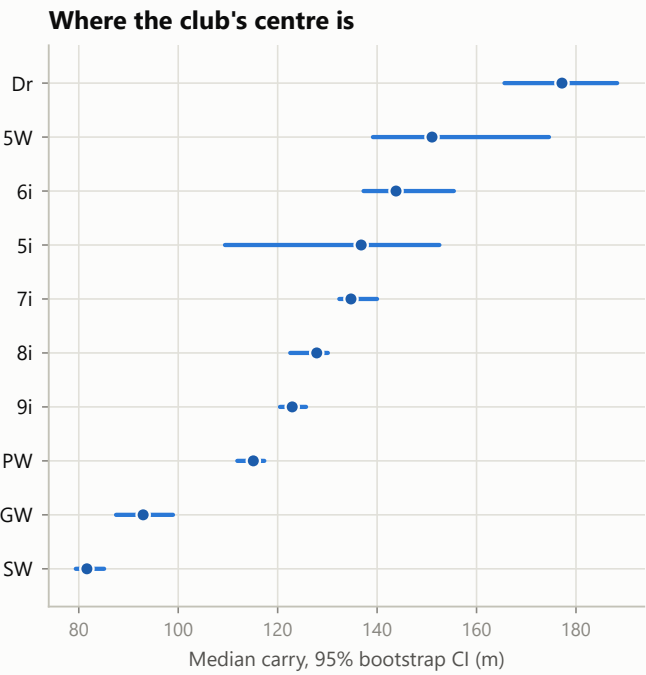
What the departures mean

The 8 iron fails hardest, and it fails on one shot: skew -2.22 with excess kurtosis $+9.62$ is the signature of a single extreme point, not of a wide body. Its $s/\text{MAD-s}$ ratio of 1.30 says the same thing from the other side - drop the tail and it is a ± 11 m club, keep it and it is a ± 14 m club. Both are true and they answer different questions: use the robust number to judge your striking, the classical one to decide whether to take on a carry.

The driver's mild left skew (-0.53) is the ordinary shape of a full-swing distribution and does not fail the test. The 5 iron's flat kurtosis (-1.04) is not a heavy tail either - it is the signature of two sessions sitting at different means, which page 10 confirms.

How much of this is real

Every number on page 3 is an estimate from a handful of swings. These are their 95% bootstrap intervals. Where an interval is wide, the club has not been measured - it has been glimpsed.



The centre is well pinned; the spread is not

Median carry is known to within ± 3 m on the well-sampled clubs (7i, 9i, PW) and to ± 20 m on the 5 iron. But the SD intervals on the right overlap almost universally: on this data you cannot claim the 6 iron is a tighter club than the 8 iron, only that the 5 iron is looser than everything. Spread needs far more shots than centre does - the table below is why.

What a sample of n buys you

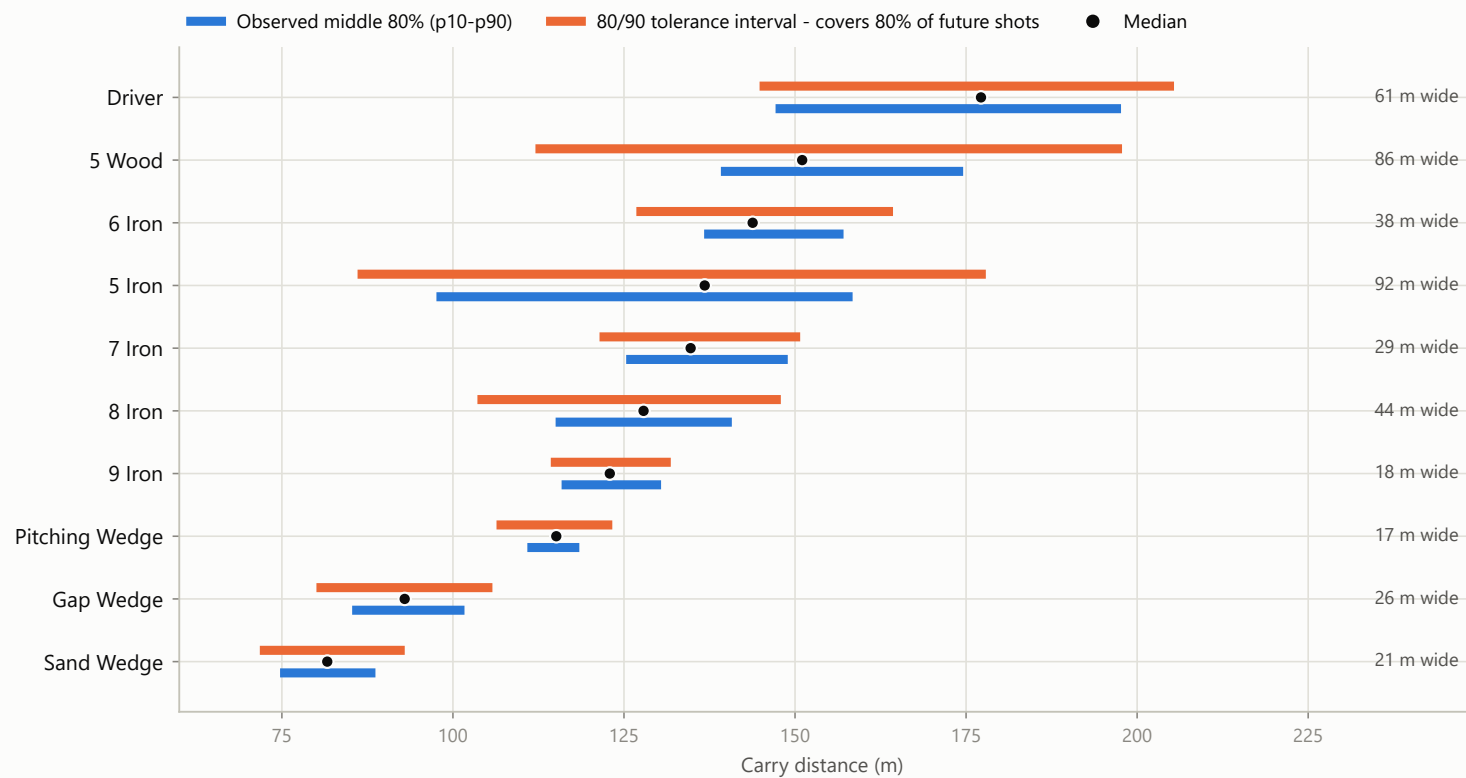
SHOTS n	CLUBS AT THIS n	95% CI FOR σ , AS A MULTIPLE OF THE OBSERVED s	WIDTH	VERDICT
n = 6	5W	0.62 $\times$ to 2.45 $\times$	3.9 $\times$	x spread essentially unmeasured
n = 8	6i, PW	0.66 $\times$ to 2.04 $\times$	3.1 $\times$	x spread essentially unmeasured
n = 13	GW	0.72 $\times$ to 1.65 $\times$	2.3 $\times$	x spread indicative only
n = 17	5i, SW	0.74 $\times$ to 1.52 $\times$	2.0 $\times$	x spread indicative only
n = 26	9i	0.78 $\times$ to 1.38 $\times$	1.8 $\times$	! spread usable for ranking clubs
n = 31	7i	0.80 $\times$ to 1.34 $\times$	1.7 $\times$	! spread usable for ranking clubs
n = 35	8i	0.81 $\times$ to 1.31 $\times$	1.6 $\times$	! spread usable for ranking clubs
n = 50	—	0.84 $\times$ to 1.25 $\times$	1.5 $\times$	+ spread usable as a planning number
n = 75	—	0.86 $\times$ to 1.19 $\times$	1.4 $\times$	+ spread usable as a planning number
n = 100	—	0.88 $\times$ to 1.16 $\times$	1.3 $\times$	+ spread usable as a planning number

The practical consequence

To state a club's dispersion to within $\pm 20\%$ - tight enough to track whether practice is working - you need roughly 50 strikes of that club, gathered across at least three sessions so the day effect sits inside the estimate rather than confounded with it. Only the 8 iron (35) and 7 iron (31) come close. A hundred-ball session spread over ten clubs measures nothing about any of them; the same hundred balls over two clubs measures both properly. If you take one number from this page: rotate a two-club block rather than a ten-club buffet, and five sessions will give you publishable dispersion on the whole bag.

The number to club off

A median is what a club does on average; it is not what to aim with. These are prediction windows - the observed middle 80% of shots, and the interval that covers 80% of future shots with 90% confidence given how little data stands behind it.



The two intervals disagree, and the gap is the sample size

For the 7 iron (n = 31) the tolerance interval is 29 m wide against an observed 24 m - a modest premium for uncertainty. For the 5 wood (n = 6) it is 86 m wide against an observed 35 m. That is not a wild club; it is a club about which almost nothing is known. Quote the blue bar when describing what has happened, and the orange bar when committing to a carry over trouble.

Carry-over-hazard numbers

The 10th percentile is the honest answer to "will it clear?" - it fails one shot in ten; the 5th is the number for water.

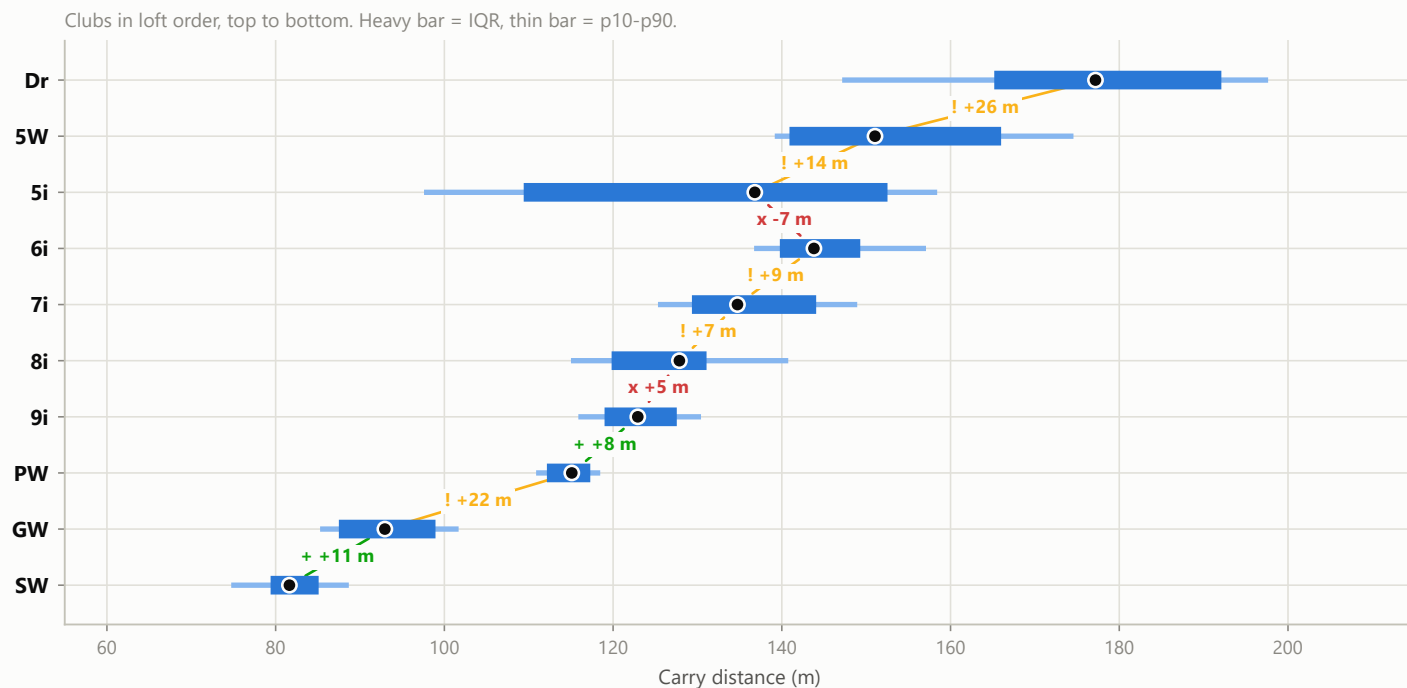
CLUB	N	MEDIAN CARRY	p25	p10 · CLEARS 9 IN 10	p05 · CLEARS 19 IN 20	SHORTEST LOGGED	TOLERANCE FLOOR	SAFETY MARGIN
Driver	29	177 m	165	147 m	141 m	133 m	145 m	x -30 m
5 Wood	6	151 m	141	139 m	139 m	138 m	112 m	! -12 m
5 Iron	17	137 m	109	98 m	90 m	81 m	86 m	x -39 m
6 Iron	8	144 m	140	137 m	136 m	135 m	127 m	+ -7 m
7 Iron	31	135 m	129	125 m	121 m	119 m	121 m	+ -9 m
8 Iron	35	128 m	120	115 m	112 m	64 m	104 m	! -13 m
9 Iron	26	123 m	119	116 m	116 m	113 m	114 m	+ -7 m
Pitching Wedge	8	115 m	112	111 m	110 m	109 m	106 m	+ -4 m
Gap Wedge	13	93 m	87	85 m	83 m	80 m	80 m	+ -8 m
Sand Wedge	17	82 m	79	75 m	73 m	73 m	72 m	+ -7 m

Read the last column as your safety margin

It is the drop from median carry to the one-in-ten short shot. The pitching wedge gives up 4 m and the 9 iron 7 m: with those two you can attack a front pin. The 5 iron gives up 39 m and the driver 30 m - with those, the pin is irrelevant and the front edge is the target. Note that the margin is not proportional to distance: the 8 iron surrenders 13 m and the 6 iron 7 m, though they carry within 16 m of each other. That asymmetry is a striking problem, not a yardage one.

Gapping: does the ladder have rungs?

A gap is only real if the longer club reliably out-carries the shorter one. P(longer wins) counts that over every shot pair: at 0.50 the two clubs are interchangeable, and below about 0.75 you cannot tell them apart on the course.



STEP	GAP IN MEDIAN CARRY	95% CI ON THE GAP	P(LONGER WINS)	COHEN d	MANN-WHITNEY p	VERDICT
Dr → 5W	+26.1 m	+0 to +43	0.78	+1.07	0.0351	! marginal - shorter wins 1 in 4
5W → 5i	+14.2 m	-9 to +50	0.75	+0.93	0.0865	! marginal - shorter wins 1 in 4
5i → 6i	-7.0 m	-34 to +11	0.40	-0.60	0.4398	x inverted - longer club is shorter
6i → 7i	+9.1 m	-0 to +19	0.75	+1.03	0.0277	! marginal - shorter wins 1 in 4
7i → 8i	+6.9 m	+3 to +16	0.75	+0.84	0.0004	! marginal - shorter wins 1 in 4
8i → 9i	+4.9 m	-2 to +9	0.62	+0.23	0.1239	x not separated - one club of range
9i → PW	+7.8 m	+4 to +13	0.88	+1.60	0.0005	+ clean rung
PW → GW	+22.2 m	+15 to +29	1.00	+3.61	<0.0001	! separated, but step is oversized
GW → SW	+11.3 m	+5 to +18	0.87	+1.62	0.0007	+ clean rung

Three findings, in order of cost

- **The 5 iron / 6 iron rung is inverted**

The 5 iron's median carry is 7.0 m short of the 6 iron's and it wins only 40% of pairs. The 95% CI on the gap runs from -34 to +11 m, so the direction is not established - but nothing in the data supports the 5 iron being the longer club. Either the shaft or the lie is wrong for you, or the club is long enough that you are steering it. On this evidence it adds no reachable distance.

- **The 8 iron / 9 iron rung barely exists**

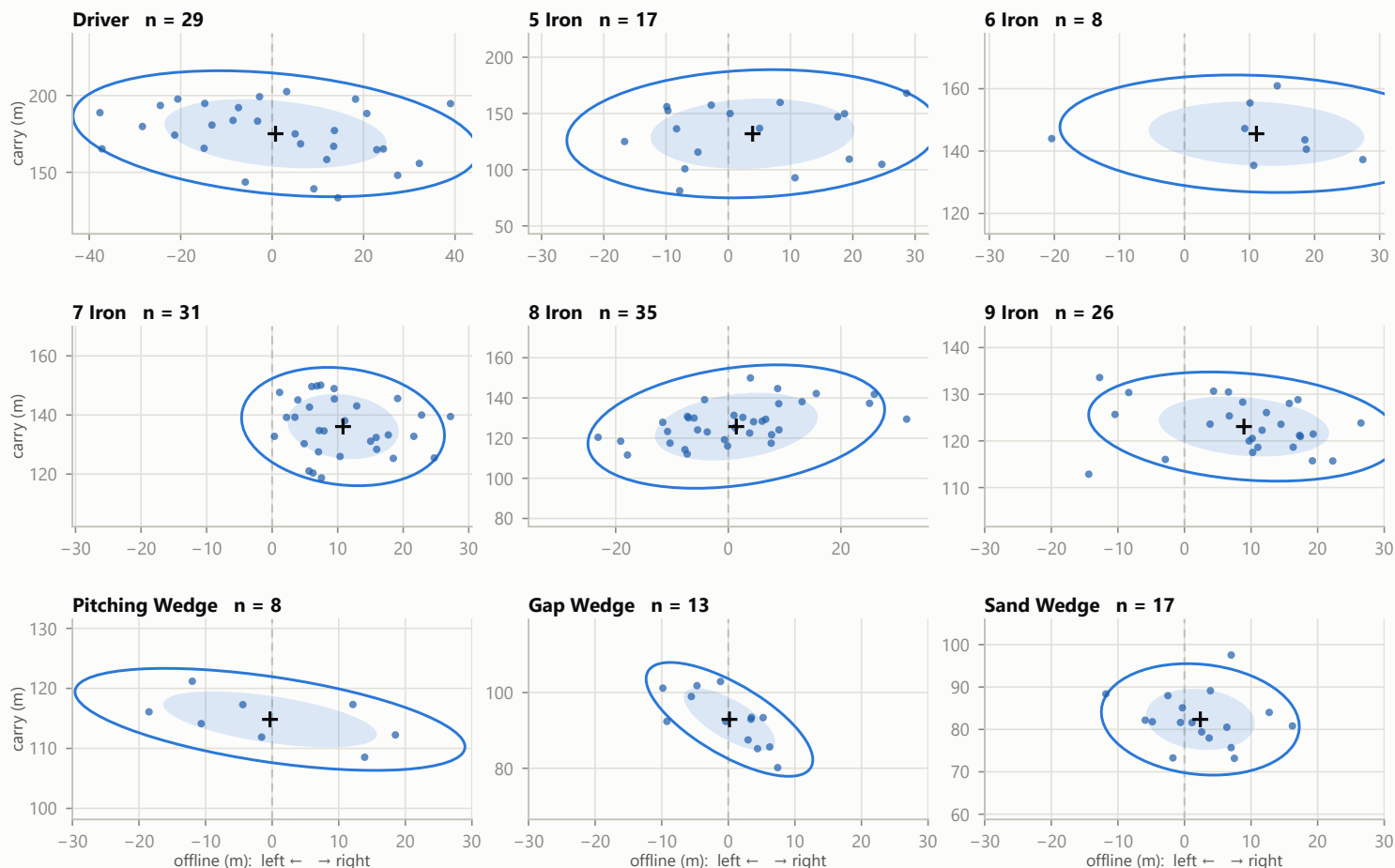
4.9 m apart, P(win) = 0.62, d = 0.23, and Mann-Whitney does not reject at p = 0.12. Two clubs are covering one club's worth of ground. On a 130 m approach the choice between them is a coin flip dressed as a decision - which is a strong argument for taking the 9 iron every time and removing the deliberation entirely.

- **Everything from the 9 iron down is clean, and the PW/GW step is too big**

9i→PW (P = 0.89), PW→GW (P = 1.00, d = 3.61) and GW→SW (P = 0.87) all separate properly. But the PW→GW step is 22 m against a bag average of 11 m: there is a hole between 93 and 115 m, exactly where a scoring approach lives. A 50° wedge, or a rehearsed three-quarter pitching wedge, fills it.

Two-dimensional dispersion

Distance error and direction error are not independent, so they are drawn together: looking down the target line, offline across and carry up. The ellipses hold 50% and 90% of the fitted bivariate normal; the dashed centre line is the target.



Directional bias, tested

CLUB	N	MEAN OFFLINE	MEDIAN	SD	ANGULAR SD	CARRY / OFFLINE r	90% AREA	WILCOXON	BIAS
Driver	29	+0.7 m	+3.2 m	20.6 m	6.8°	-0.28	0.55 ha	0.8314	+ centred
5 Wood	6	+7.6 m	+8.1 m	12.5 m	5.0°	-0.27	0.31 ha	0.2188	+ centred
5 Iron	17	+3.9 m	+0.3 m	13.9 m	6.4°	+0.11	0.53 ha	0.3060	+ centred
6 Iron	8	+11.1 m	+12.5 m	14.1 m	5.7°	-0.11	0.18 ha	0.1484	+ centred
7 Iron	31	+10.8 m	+7.9 m	7.2 m	3.1°	-0.15	0.10 ha	<0.0001	x pushed 11 m right
8 Iron	35	+1.4 m	+1.2 m	12.3 m	5.7°	+0.28	0.24 ha	0.5876	+ centred
9 Iron	26	+8.9 m	+10.6 m	10.9 m	5.1°	-0.21	0.08 ha	0.0007	x pushed 9 m right
Pitching Wedge	8	-0.3 m	-3.0 m	13.6 m	6.9°	-0.54	0.07 ha	0.9453	+ centred
Gap Wedge	13	+0.2 m	+3.0 m	5.8 m	3.6°	-0.71	0.04 ha	0.9460	+ centred
Sand Wedge	17	+2.4 m	+2.6 m	6.9 m	4.8°	-0.13	0.06 ha	0.1594	+ centred

Two things the ellipses show that the SDs hide

- The wedges tilt.

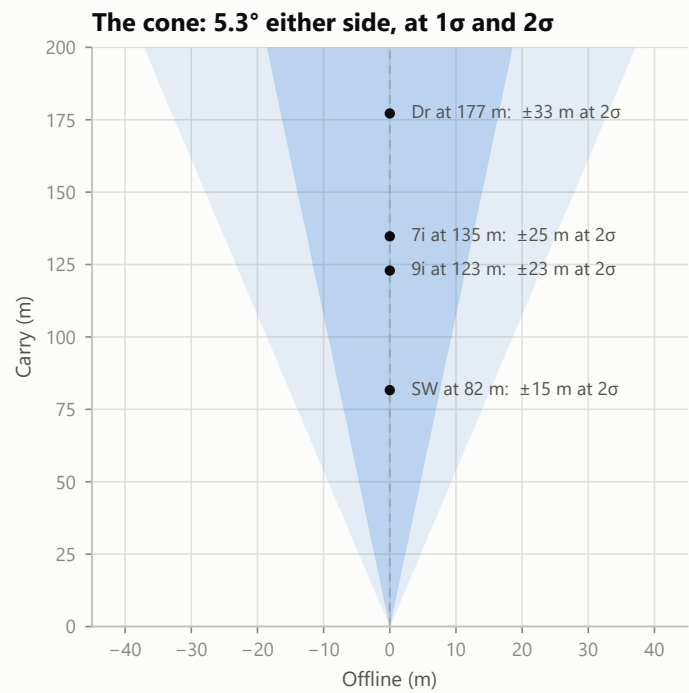
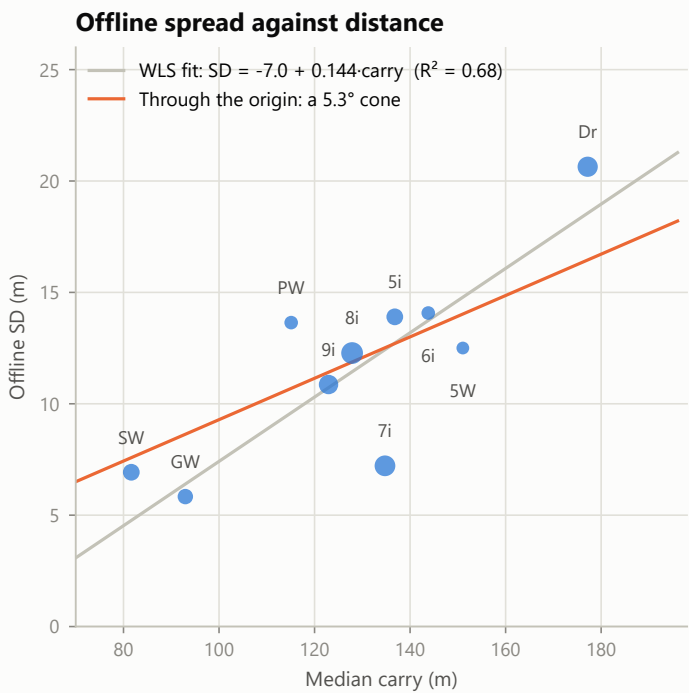
Gap wedge $r = -0.71$ and pitching wedge $r = -0.54$ between offline and carry. Your wedge miss is a single fault, not two - the ball starts right and comes up short together, which is the signature of a face open to the path with the low point behind the ball. One correction fixes both axes; treating distance control and direction as separate practice tasks wastes half the work.

- Bias and spread are different problems with different fixes.

The 7 iron has the tightest offline SD in the bag (7.2 m) and the largest push (+10.8 m): its ellipse is small and sitting in the wrong place, which is an aim or face-angle problem correctable in a session. The driver is the reverse - mean offline +0.8 m, SD 20.6 m. Perfectly aimed, and 41 m wide at two sigma.

Does the miss scale with the club?

If direction error were a constant angle, offline spread would rise in proportion to carry and the whole bag would sit on one line through the origin. It nearly does - which means dispersion is one property of your swing rather than ten properties of ten clubs.



The fit, and the two clubs that break it

A weighted least-squares fit (weights $n-1$) explains 68% of the variation in offline SD across the bag, with a slope of 0.144 m of spread per metre of carry (95% CI 0.064 to 0.225, $p = 0.003$). Forced through the origin the slope becomes a single angle: 5.3°. That is the number to carry into course strategy, because it converts any yardage into a width.

CLUB	MEDIAN CARRY	OFFLINE SD	PREDICTED BY THE FIT	RESIDUAL	2σ WIDTH ON THE GROUND	READ AS
Driver	177 m	20.6 m	18.6 m	+2.1	83 m	· on the line
5 Wood	151 m	12.5 m	14.8 m	-2.3	50 m	· on the line
5 Iron	137 m	13.9 m	12.7 m	+1.2	56 m	· on the line
6 Iron	144 m	14.1 m	13.7 m	+0.3	56 m	· on the line
7 Iron	135 m	7.2 m	12.4 m	-5.2	29 m	+ tighter than the bag average
8 Iron	128 m	12.3 m	11.4 m	+0.8	49 m	· on the line
9 Iron	123 m	10.9 m	10.7 m	+0.1	43 m	· on the line
Pitching Wedge	115 m	13.6 m	9.6 m	+4.0	55 m	x looser than the bag average
Gap Wedge	93 m	5.8 m	6.4 m	-0.6	23 m	· on the line
Sand Wedge	82 m	6.9 m	4.8 m	+2.2	28 m	· on the line

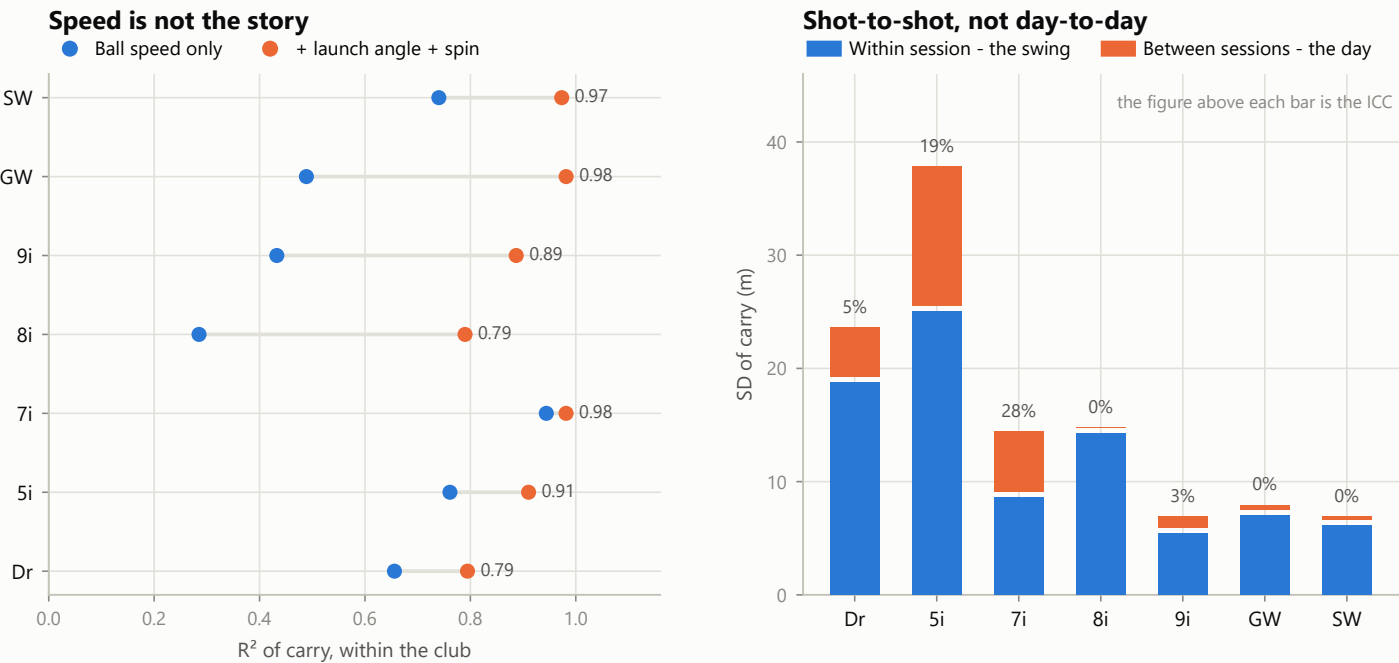
What a 5.3-degree cone costs on the course

At driver distance the cone is 33 m wide at one sigma and 66 m at two. A 40 m fairway is held about three-quarters of the time on the cone, and about two-thirds using the driver's own measured spread - even when nothing goes wrong. That is not a swing fault, it is the arithmetic of the club, and it is the case for hitting less club off a tight tee: the cone narrows in proportion, so trading driver for 5 wood buys back a third of the width at a cost of 26 m. The 5 wood's own dispersion is unmeasured at $n = 6$, so test that before trusting it.

The 7 iron sits 5 m below the line and the pitching wedge 4 m above it. The 7 iron's advantage is the one genuinely well-established dispersion result in this report ($n = 31$): whatever you are doing with that club is worth generalising. The pitching wedge's excess rests on 8 shots and should not be acted on yet.

Where the variance comes from

Two decompositions. Left: how much of each club's carry spread is explained by ball speed alone, against ball speed plus launch angle and spin. Right: how much of it is the day, and how much is the swing. Both point the same way.



Clubhead speed explains almost none of your scoring-iron spread

For the 8 iron, ball speed alone accounts for $R^2 = 0.29$ of the carry variation; adding launch angle and spin takes it to 0.79. The 9 iron goes 0.43 → 0.89 and the gap wedge 0.49 → 0.98. In plain terms: when your 8 iron comes up short it is usually not because you swung slower - it is because the ball came off at the wrong angle with the wrong spin. That is a strike-quality and low-point problem, and it responds to strike drills rather than speed work. The driver and 7 iron are the exceptions (0.66 and 0.94 from speed alone), where distance genuinely does track how hard you hit it.

Variance components by club

CLUB	SESSIONS	SD WITHIN SESSION	SD BETWEEN SESSIONS	ICC	TOTAL SD	INTERPRETATION
Driver	3	18.8 m	4.3 m	0.051	19.3 m	! mild day effect
5 Wood	1	17.6 m	—	—	17.6 m	· one session - no day effect estimable
5 Iron	2	25.1 m	12.3 m	0.194	27.9 m	! mild day effect
6 Iron	1	8.8 m	—	—	8.8 m	· one session - no day effect estimable
7 Iron	2	8.6 m	5.3 m	0.279	10.1 m	x drifts between days - average sessions
8 Iron	2	14.3 m	0.0 m	0.000	14.3 m	+ the day barely matters; range = course
9 Iron	2	5.4 m	1.0 m	0.035	5.5 m	+ the day barely matters; range = course
Pitching Wedge	1	4.0 m	—	—	4.0 m	· one session - no day effect estimable
Gap Wedge	2	7.0 m	0.4 m	0.003	7.0 m	+ the day barely matters; range = course
Sand Wedge	2	6.1 m	0.3 m	0.002	6.1 m	+ the day barely matters; range = course

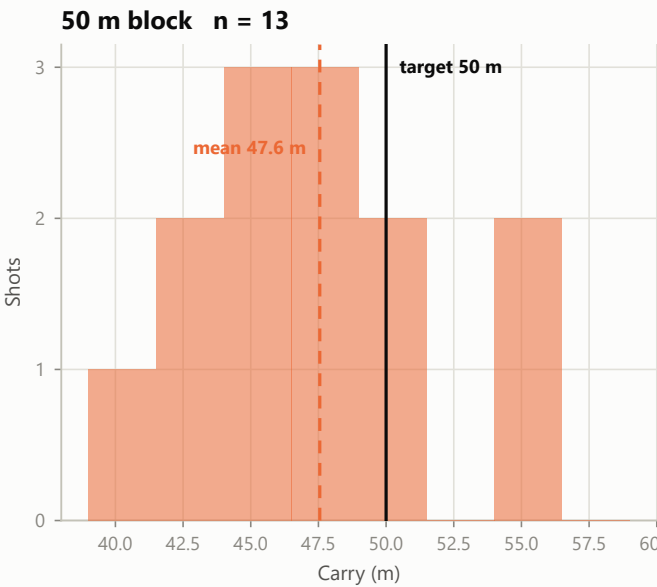
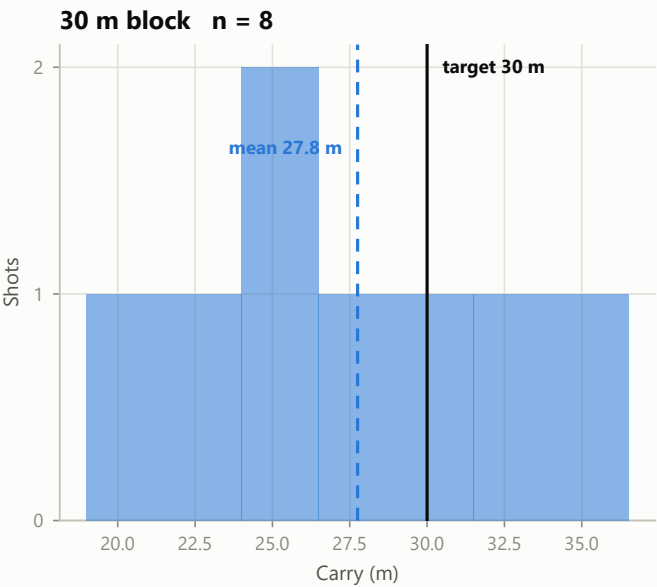
The 5 iron and 7 iron are the ones that drift

ICC = 0.19 for the 5 iron and 0.28 for the 7 iron, against under 0.05 for everything else. The 5 iron's two sessions sat 39 m apart in median carry - 109 m on 11 April, 148 m on 16 April. That is not a club with a wide distribution, it is a club you have not settled on a method with. Averaging those two sessions produces a yardage that was never hit on either day, which is exactly the failure mode a variance-components model exists to catch.

The counterpart is the good news: for five of seven repeated clubs, less than 5% of the carry variance is between-session. Your range numbers are not a range artefact - they will hold on the course, and the independent on-course data on page 12 confirms it.

Distance control: the lob wedge blocks

Both lob-wedge blocks were hit to a nominated carry - 30 m and 50 m - which makes them the only part of this data set with a known target. That converts dispersion into something a distribution alone cannot give you: accuracy, separable from precision.



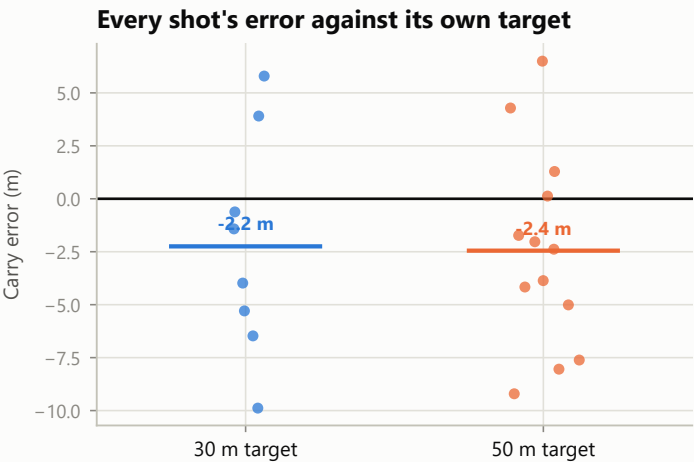
Both blocks land short; neither is significant on its own

BLOCK	N	TARGET	MEAN CARRY	BIAS	BIAS %	SD	CV	95% CI ON BIAS	t TEST p	VERDICT
30 m	8	30.0 m	27.8 m	-2.2 m	-7.5%	5.27 m	19.0%	-6.7 to +2.2	0.267	! short, not significant
50 m	13	50.0 m	47.6 m	-2.4 m	-4.9%	4.67 m	9.8%	-5.3 to +0.4	0.083	! short, not significant

Precision improves with length; accuracy does not

The 30 m block carries a CV of 19.0% against 9.8% at 50 m - roughly double the relative scatter at the shorter target, from a smaller absolute SD (5.27 m against 4.67 m). This is the standard finding for partial wedges and it has a mechanical cause: at 30 m you are further down the speed range, where a given error in clubhead speed or strike location is a larger fraction of the total energy. It is also why the shortest wedge shots are the hardest to control, and why a rehearsed clock system beats feel below about 40 m.

The bias is the more useful finding. Both blocks land short by nearly the same fraction - -7.5% at 30 m and -4.9% at 50 m - so this is a proportional calibration error rather than a fixed offset. Neither is individually significant at these sample sizes, but a proportional error agreeing in sign across two independent targets is worth a note: your internal yardage for the lob wedge reads about 5-7% long. Recalibrating the number is free; correcting a swing that is not broken is not.



What is missing

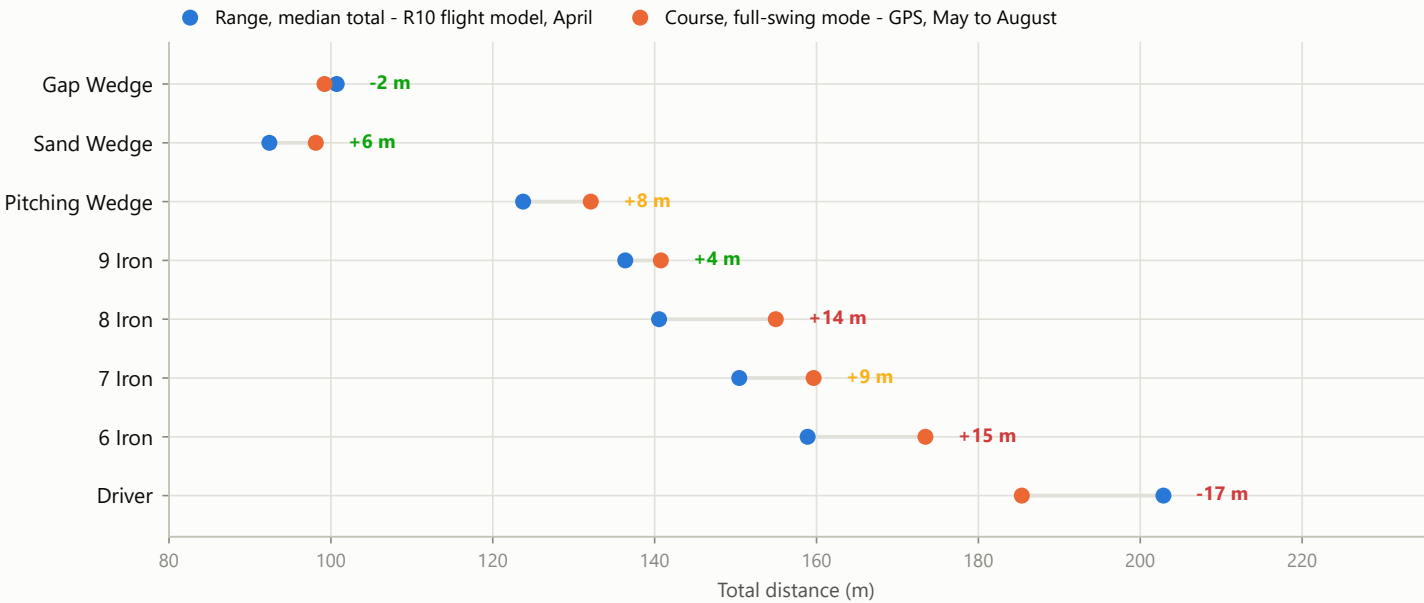
The rest of the bag has no target attached, so only precision can be measured - never accuracy. A club can be beautifully repeatable and 8 m off the number written on your yardage card, and nothing in pages 3 to 9 would reveal it.

The fix costs nothing: nominate a carry before each block and log it in the Note field, as these two blocks already do. Ten shots at a nominated number per club turns this report from a description of your dispersion into a calibration of your yardage card - which is the version that saves shots.

It also makes the wedge matrix possible: three clubs at three rehearsed lengths gives nine calibrated carries between 30 and 115 m, which would close the PW-to-GW hole found on page 7 without adding a club.

Does it hold up on the course?

An independent check from 598 GPS-tracked strokes over 14 rounds, May to August 2026. Course shots mix full swings with layups and chips, so each club's lengths are split by a two-component Gaussian mixture and only the upper component is compared.



The irons play longer outside; the driver plays shorter

CLUB	RANGE TOTAL	COURSE SHOTS	FULL SWINGS	FULL-SWING MODE	SD	PARTIAL MODE	DELTA	LIKELY CAUSE
Driver	203 m	41	73%	185 m	28.5 m	30 m	-17 m	a range block hides misses
6 Iron	159 m	21	68%	173 m	19.2 m	132 m	+15 m	summer roll and warmer air
7 Iron	150 m	53	76%	160 m	19.2 m	84 m	+9 m	summer roll and warmer air
8 Iron	141 m	75	70%	155 m	17.8 m	80 m	+14 m	summer roll and warmer air
9 Iron	136 m	47	75%	141 m	14.6 m	59 m	+4 m	agrees inside noise
Pitching Wedge	124 m	48	83%	132 m	14.3 m	70 m	+8 m	summer roll and warmer air
Sand Wedge	92 m	39	70%	98 m	10.5 m	62 m	+6 m	agrees inside noise
Gap Wedge	101 m	46	52%	99 m	12.3 m	26 m	-2 m	agrees inside noise

Reading the disagreement

The mid-irons run 4 to 15 m longer on the course than the R10's total for the same club, and three causes pull the same way without the data being able to separate them: the rounds were played in May-August air (roughly 1.16 g/L against the range's 1.19-1.23, worth about +2%), summer fairways give more roll than range turf, and the GPS length is measured ball-to-ball on the ground, which credits every metre of bounce. The direction and rough size are what matter - add about 8 m to the range total for a mid-iron in summer, and nothing in winter.

The driver is the honest one. It plays 17 m shorter on the course and its course SD (28.5 m) is half again its range SD (19.1 m). A range block is a best case - same lie, same ball, no consequence, no first tee. That gap is not measurement error; it is what happens when the shot counts, and it is the strongest argument in this report for planning off the p10 column on page 6 rather than the median.

Lie matters more than club

CLUB AND LIE	SHOTS	MEDIAN DISTANCE	INTERQUARTILE RANGE	IQR WIDTH
7 Iron from the fairway	10	152 m	144 - 157 m	13 m
7 Iron from the tee	25	152 m	128 - 172 m	45 m
7 Iron from the rough	18	147 m	103 - 166 m	63 m
8 Iron from the fairway	20	151 m	134 - 165 m	31 m
8 Iron from the tee	34	149 m	117 - 160 m	43 m
8 Iron from the rough	20	121 m	74 - 159 m	85 m

Course distances are start-to-end GPS separations from Shot Scope - total including roll, and including every mishit. The mixture is fitted per club with two components and ten restarts; the reported mode is the longer component. The gap wedge is logged on the course as a 52-degree wedge.

What to do about it

Findings ranked by expected strokes saved, each with its statistical warrant and what would overturn it.

1 Take the 5 iron out of the bag until it is refit

It carries 7 m shorter than the 6 iron with three times the spread ($s = 26.6$ m) and wins 40% of head-to-head pairs. Its 10th percentile is 98 m - a club you cannot commit to. Warrant: $n = 17$ across two sessions; the 95% CI on the gap (-34 to +11 m) does not establish the inversion, but it firmly excludes the 5 iron being usefully longer. Overturned by: 20 shots in a single settled session with a median above 150 m. Until then a 5 wood or a hybrid covers 150-160 m better.

2 Fix the aim before the swing on the 7 and 9 irons

Mean offline +10.8 m (7i) and +8.9 m (9i), both $p < 0.001$, sitting on top of the smallest offline SD in the bag for the 7 iron. A biased-but-tight club is the cheapest fix in golf: the pattern is already repeatable, it is simply pointed wrong. Check alignment and face angle at address first - the club path data (mean +2.5° to +5.7° with the face open to it) says the ball is starting right rather than curving there. Warrant: $n = 31$ and $n = 26$, significant under both t and Wilcoxon, and present in both sessions for each club.

3 Practise strike quality, not clubhead speed, in the scoring irons

Ball speed alone explains only 29% (8i), 43% (9i) and 49% (GW) of within-club carry variation; adding launch and spin takes those to 0.79, 0.89 and 0.98. Your short shots are launch-and-spin failures, which means low point and face contact. Warrant: OLS within club, $n \geq 13$. This is also what produced the 8 iron's 63.9 m outlier and its -2.22 skew.

4 Fill the 93-115 m hole, and stop deliberating between 8 and 9 iron

PW→GW is a 22 m step against a bag average of 11 m, sitting exactly in scoring range. Meanwhile 8i→9i is 4.9 m with $P(\text{win}) = 0.62$ - two clubs doing one club's job. A 50° wedge, or a rehearsed three-quarter pitching wedge calibrated the way page 11's blocks were, closes the hole; defaulting to the 9 iron in the overlap removes a decision the data says is a coin flip.

5 Recalibrate the lob wedge yardage down by about 6%

Both nominated blocks landed short by nearly the same fraction (-7.5% at 30 m, -4.9% at 50 m). Neither is individually significant, but a proportional error agreeing in sign across two targets is worth a card change rather than a swing change. Warrant: weak - $n = 8$ and $n = 13$ in one session. Confirm with a second nominated block before acting.

6 Plan off the 10th percentile, not the median

The driver plays 17 m shorter on the course than on the range and half again as wide (SD 28.5 m against 19.1 m). The median is what a club does in a range bay; the p10 column on page 6 is what it does when the shot counts. For any carry over trouble, that column is the number.

The measurement plan that would make the next report definitive

Half the conclusions above are limited by n , not by analysis. Five clubs have fewer than 18 strikes, and a spread estimated from eight shots is uncertain by a factor of three. Two changes to how the data is collected would remove that limit entirely:

- Two clubs per session, not ten. Fifty strikes of one club across three sessions pins its dispersion to $\pm 20\%$ and separates the day effect from the swing. Ten clubs at ten balls each measures nothing about any of them. Five sessions on this rotation covers the bag.
- Nominate a carry and log it. The lob-wedge blocks are the only shots here with a target, and they are the only shots that could be tested for accuracy rather than merely described for precision. One line in the Note field turns every block into a calibration.

With those two changes the same script produces tolerance intervals narrow enough to build a yardage card on, ICCs that are point estimates rather than ranks, and a wedge matrix of nine calibrated carries. The analysis is not the binding constraint here - the sample is.

FULL-SWING STRIKES

190

over 10 clubs, plus
21 partial wedges

CLUB FAILING NORMALITY

1

the 8 iron, on a single
thinned strike

RUNGS BROKEN

2

plus 4 more marginal,
of nine steps

SHOTS PER CLUB NEEDED

~50

to state dispersion
to within $\pm 20\%$